

Amplified linear and nonlinear chiral sensing assisted by anapole modes in hybrid metasurfaces

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Supplementary Material

SI. SCATTERING CROSS-SECTIONS OF A SILICON HOLED DISK IN WATER

The results of the multipole decomposition analysis carried out for the results shown in Figure 1 of the main document are shown in Figure S1 and corresponding to the 0.1 steps of the interval $\frac{h}{R_{Si}} \in [0.5, 1.7]$. An almost perfect fit to the FDTD results is achieved upon consideration of multipoles up to octupole order. The categorization of resonances in Figure 1a is based on peak recognition for ED, MD, EQ and MQ resonances, while anapolar modes (EDAs, MDAs and EQA) were recognized as valleys.

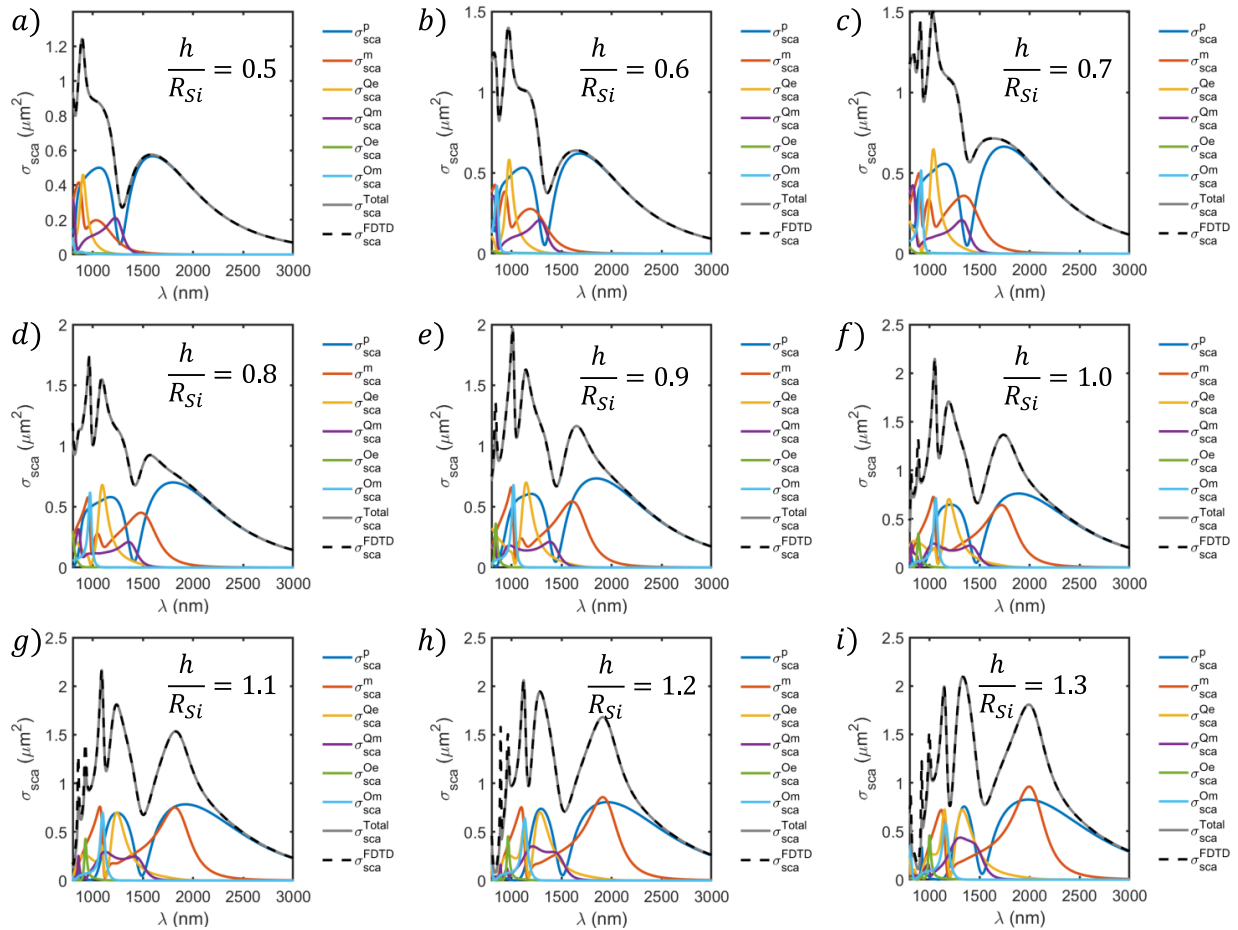


Figure S1. Scattering cross-section with multipole decomposition for silicon holed disks with aspect ratios a) 0.5 to m) 1.7.

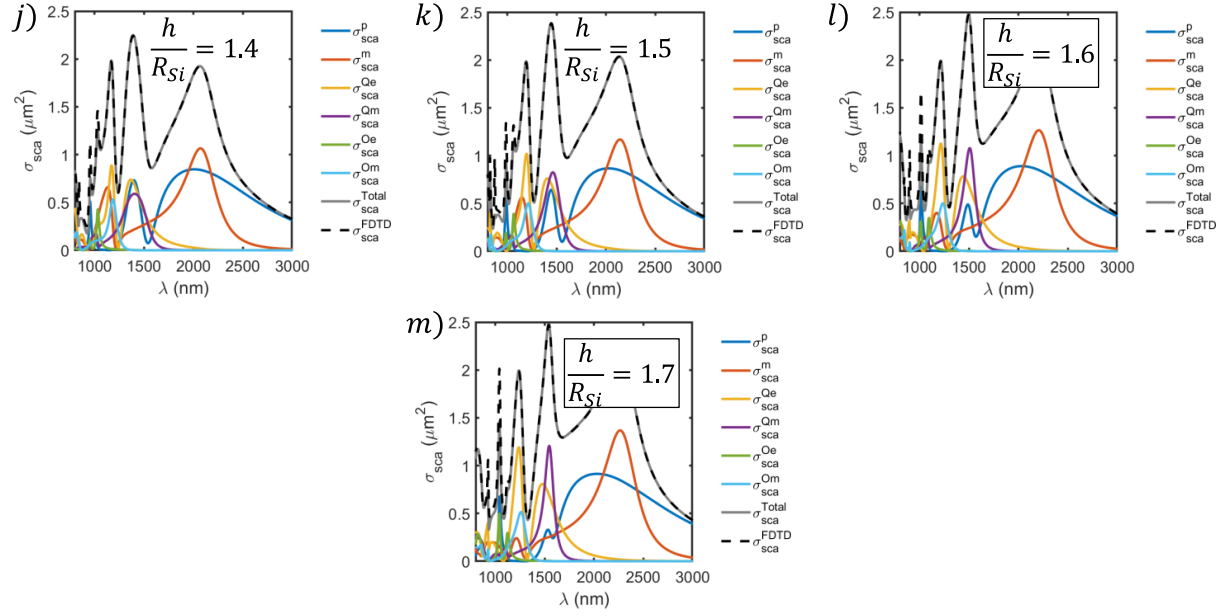


Figure S1 (cont.). Scattering cross-section with multipole decomposition for silicon holed disks with aspect ratios a) 0.5 to m) 1.7.

SII. INFLUENCE OF THE ALUMINA SUBSTRATE ON THE AVERAGE OPTICAL CHIRALITY DENSITY

The results for averaged optical chirality density, electric field, magnetic field and phase factor are summarized in Figure S2 in a similar format than Figure 1 of the main text. We observe that, in general, fields are less intense, leading to a lower average chirality within the disk gap. However, all trends remain the same and the phase factor is not affected in any way.

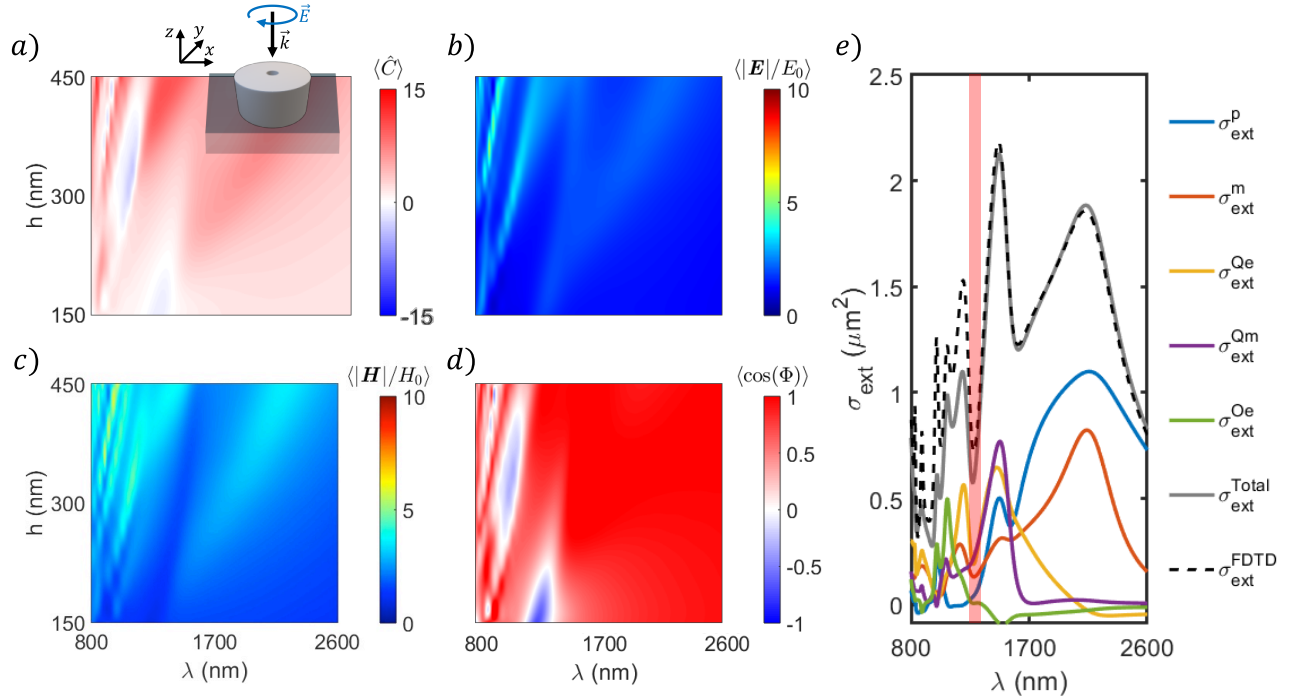


Figure S2. Evolution of the a) optical chirality density, b) electric field, and c) magnetic field enhancements and d) phase factor $\cos(\Phi)$ averaged within the inner gap of the disk-substrate system with respect to the disk height and the incident wavelength λ . e) Extinction cross-section for the $h = 450$ nm case with multipole decomposition up to the electric octupole term. In red, the spectral position of point A, where anapole resonances overlap, is highlighted.

This is also seen in the far field multipole decomposition of the extinction cross-section (Fig. S2e), obtained following work by Evlyukhin and coworkers¹. In this case, precision in comparison to FDTD results is lower as a substrate introduces cross-talk between multipoles², meaning that the considered multipoles are not the true eigenmodes of the system¹. This also explains the fact that some multipoles have a negative contribution into the cross-section. Despite this, the same trends are generally observed, as the substrate has a relatively low refractive index contrast with water. As often happens with substrates, the electric dipole contribution is widened due to the mode profile extending into the substrate¹. A notorious dark mode near $\lambda = 1300$ nm is found as a result of several anapole resonances (mainly magnetic dipole and electric quadrupole) overlapping.

SIII. GEOMETRICAL OPTIMIZATION OF THE GOLD RING

As mentioned in the main text, the optimization process consisted of two steps. In the first one, the inner radius was fixed to the minimum considered value ($r_{Au} = 350$ nm), while the outer radius was varied until an optimum point was found. The results for this first step, together with more details, are found in Figure S3.

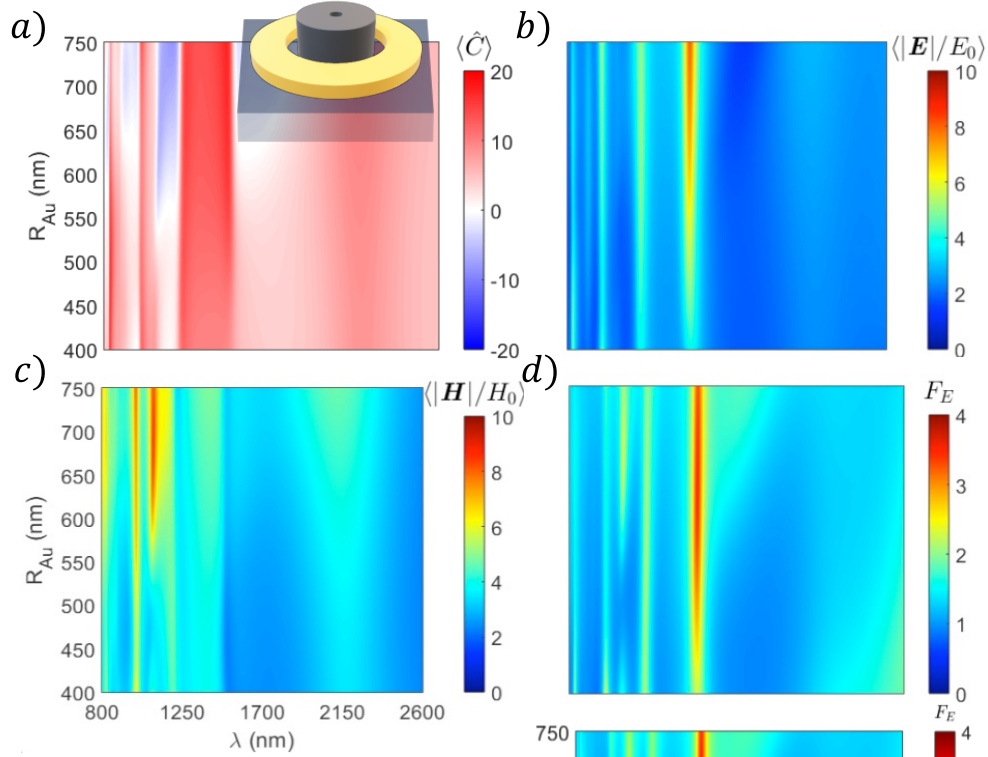


Figure S3. FDTD results obtained for the optimization of the gold ring. a) Results for the optical chirality density, b) normalized electric field intensity F_E inside the silicon disk, c) electric field and d) magnetic field enhancements; for several outer radii of the gold ring, with a fixed inner radius $r_{Au} = 350$ nm.

Figures S3a and S3d are equivalent to Figure 2a and 2b in the main text and are repeated here for reference. Focusing on the extra information provided by this figure, we can see that near the chirality enhancing broadband, two narrow lines with intense local electric fields appear (Figure S3b). These correspond to the EDA mode (around 1500 nm) and the superposition of modes discussed in the previous section (around 1220 nm). This is further confirmed by the magnetic field map (Figure S3c), where we find a more broadband amplification, contributed by the MQ mode (near the EDA resonance); and a narrow line corresponding to the MDA mode near 1200 nm. In general, all fields seem to have larger intensifications for bigger gold rings (larger radii).

Since the OCD average in the 1230-1500 nm band does not vary significantly with respect to R_{Au} , and the F_E factor reaches its maximum at $R_{Au} = 650$ nm, this value was fixed for the second step of the optimization process, where the inner radius was varied. The results for this step are found in Figure S4. Again, Figures S4a and S4d are equivalent to Figure 2c and 2d of the main text. In general, this figure displays analogue results to the previous one. The same trends found in the first step are observed, though reversed, suggesting that the overall field concentrations are correlated with the amount of gold in the structure. Focusing again in the F_E factor, in Figure S4d, we observe an overall maximum at $r_{Au} = 400$ nm.

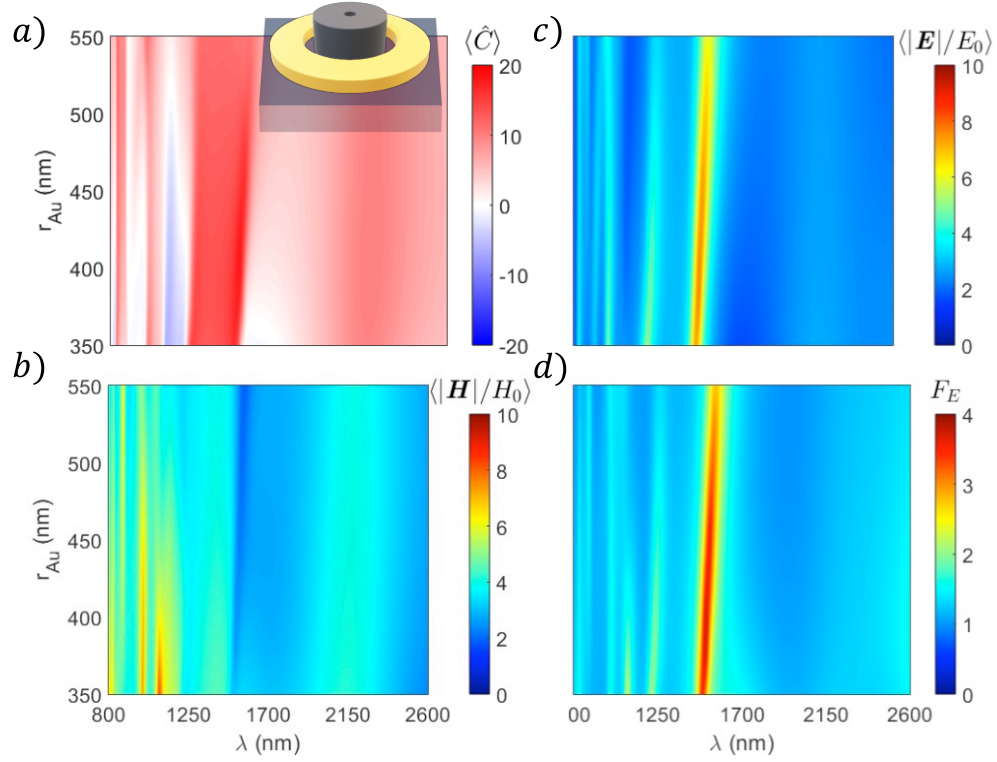


Figure S4. FDTD results obtained for the optimization of the gold ring. a) Results for the optical chirality density, b) normalized electric field intensity F_E inside the silicon disk, c) electric field and d) magnetic field enhancements; for several inner radii of the gold ring, with a fixed outer radius $R_{Au} = 650$ nm.

To obtain a better understanding of the electromagnetic response, we perform a multipole decomposition of the gold-ring system optimal geometry in an homogeneous water medium. As demonstrated in Figure S2e, the addition of the substrate does not significantly change the overall response of the system in comparison to the homogeneous result in Figure 1e of the main text. However, the addition of the substrate causes a significant limitation to the precision of the multipole decomposition method. Therefore, the substrate has been eliminated from this calculation for the sake of clarity.

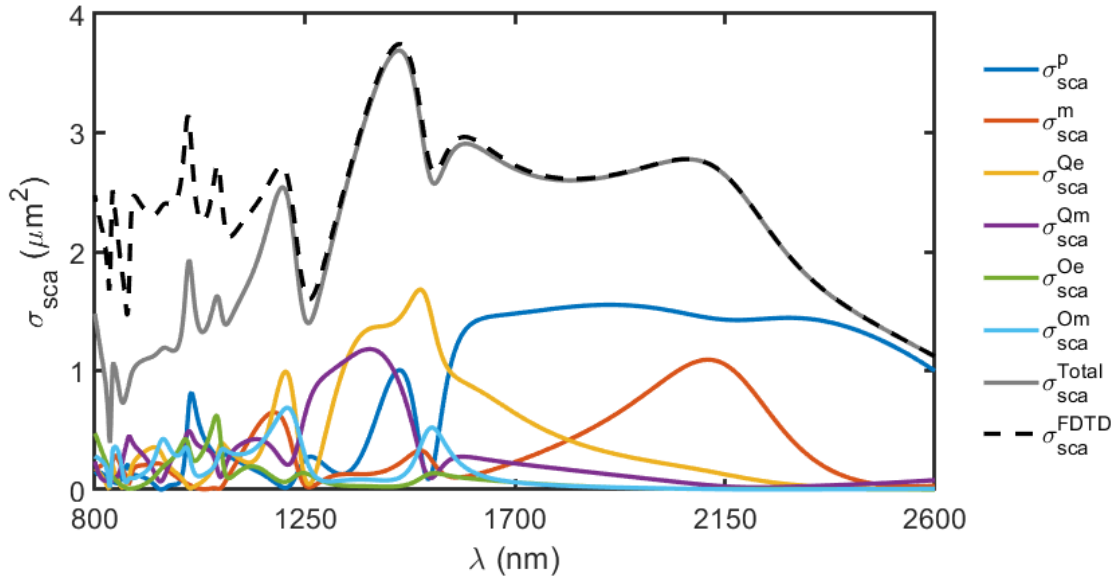


Figure S5. Far field scattering cross-section for the $(R_{Au}, r_{Au}) = (650, 400)$ nm system with multipole decomposition up to the magnetic octupole term.

The results for this analysis are shown in Figure S5. Here we can see that a good fit with FDTD is obtained for wavelengths longer than 1200 nm. A very broad electric dipolar response is seen at the longer wavelengths, owing to the well-known broad response of gold rings^{3,4}. The introduction of the ring also redshifts the excitation of higher order terms significantly, as the electric quadrupole displays a dual maximum at 1400 and 1500 nm. The magnetic octupole resonance is also redshifted to 1500 nm. This general redshift also explains the mismatch with FDTD at shorter wavelengths since contributions from higher terms become important. Despite all this, the dark modes are still observed, near 1250 (main contributions from magnetic dipole and electric quadrupole terms) and 1500 nm (electric dipole contribution) respectively. These agree well with the results in Figures S3 and S4, especially considering the spectral position of F_E maxima; and support the idea that an efficient THG is obtained due to anapole resonances.

SIV. DETAILS OF THERMAL AND NONLINEAR CALCULATIONS

In order to obtain precise results for thermal calculations, we use the RF module of the COMSOL Multiphysics suite since it provides with a built-in strategy to solve both light-matter interaction and thermal dissipation problems in an integrated manner. To keep consistency with the previously conducted FDTD simulations, material data was extracted and interpolated from the same sources.

Analogously to non-linear calculations, the light-matter interaction problem under CW illumination was solved in two steps by means of FEM. Initially, Maxwell's equations are solved to determine the background electric field (plane wave) within the simulation region in absence of the plasmonic structure. Subsequently, these equations are solved again, this time considering the nanostructure boundary conditions. This step provides with the spatial distribution of the scattered electric field, the scattering and absorption cross sections, and the absorbed/emitted electromagnetic power, among others. The second step involves solving the heat equation, based on the previous electromagnetic analysis. Particularly, the absorbed electromagnetic power (resistive losses) is used to feed the thermal calculations which finally allows for obtaining the spatial temperature fields. A convective node across the simulation region boundaries was used to enable the cooling of the whole system.

The near field spatial distributions of TH electric field and temperature are shown in Figure S6. Note that, once normalized, the temperature distribution is the same for all wavelengths. We observe in Figures S6a-b, that the TH field suggests transmission for 1220 nm incidence, while emission in the 503 nm (1510 nm incidence) wavelength is mostly side and upwards.

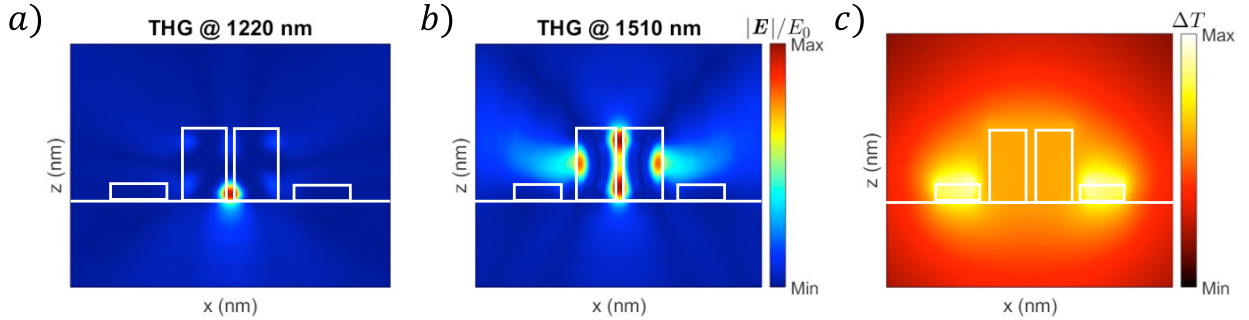


Figure S6. Normalized spatial distributions of a) TH electric field at a 406 nm wavelength (1220 nm incidence), b) TH field at 503 nm (1510 nm incidence) and c) temperature increment.

To explore the thermal behavior of the system under pulsed illumination, we adopt the theoretical framework established by Baffou and coworkers⁵, assuming a Gaussian temporal profile with peak intensity I_0 equivalent to the intensities studied in Figure 3b; and a pulse length of 100 fs. This pulse length is sufficiently shorter than the characteristic gold electron phonon interaction time ($\tau_{e-ph} = 1.7$ ps), so that the heat power can be properly approximated as $Q(t) \approx \sigma_{abs} F p(t)$, where σ_{abs} is the wavelength-dependent absorption cross-section of the gold ring, obtained from previous CW simulations, F is the pulse fluence and $p(t) = \frac{1}{\tau_{e-ph}} \exp(-\frac{t}{\tau_{e-ph}})$ the time-dependent power transfer function from the electron gas to the atomic lattice.

Interfacial Thermal Conductances (ITCs) were considered at the gold-water (200 MW/m²·K⁶) and silicon-water (143 MW/m²·K⁷) interfaces.

FEM time-dependent heat transfer simulations were carried out in COMSOL, covering a whole microsecond, which is the time between pulses for an assumed repetition rate of 1 MHz. To show that this repetition rate is sufficient to allow thermal dissipation between pulses, Figure S7 contains the temporal evolution of temperature in both the gold ring and the inner gap of the silicon disk.

It can be seen that the thermal response of the structure is not immediate and temperature increments take around 100 fs to rise, reaching maximum values around the picosecond range and then fall, vanishing at $t_{rep} = 1 \mu s$, meaning that for the assumed pulse repetition rate $1/t_{rep} = 10^6$ Hz the structure will not accumulate heat between pulses. For the inner gap of the structure, where samples are intended to be located, this growth and decay shape of the temperature increment is significantly delayed, due to the heat transfer from the gold ring and the overall lower thermal conductivity of water compared to that of the alumina substrate. Likewise, it reaches zero values at the end of the pulse cycle suggesting negligible potential overheating of the sample.

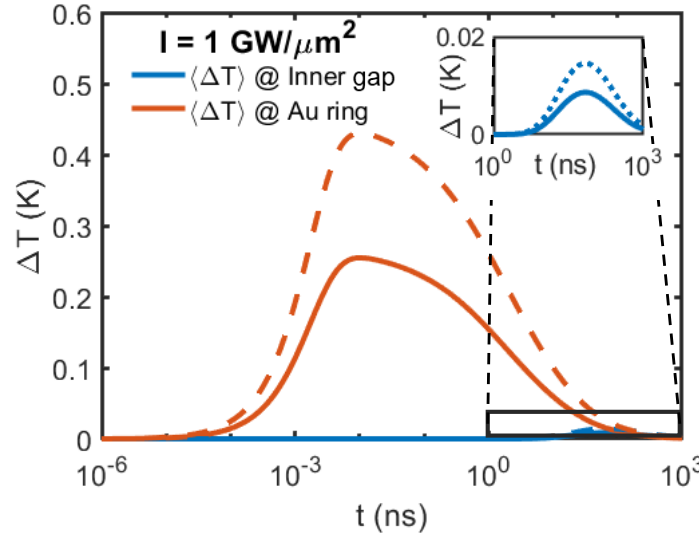


Figure S7. Time evolution of the average temperature inside the inner gap of the Si disk (blue) and inside the gold ring (red) from $t = 1$ ns to $t = 1 \mu s$ at 1220 nm (solid lines) and 1510 nm (dashed lines) wavelengths.

SV. METASURFACE OPTIMIZATION

Once the unit cell has been designed and analyzed, we propose a metasurface design to further illustrate its capabilities to enhance the CD effect. A square lattice with period Λ is arranged, and the dependence of OCD with this period is studied for the interval $\Lambda \in [1500, 3000]$ nm in steps of 100 nm. Both the transmittance and reflectance of the metasurface, as well as the average OCD inside the inner gap were calculated using FDTD methods, with the relevant results shown in Figure S8.

As shown in Figure S8a, the metasurface period does not have a significant impact on the spectral position of the high OCD band, being approximately constant in the 1230-1500 nm interval for both tightly packed ($\Lambda = 1500$ nm) and widely spaced ($\Lambda = 3000$ nm) metasurfaces, where the overall behaviour is close to that of the unit cell. This can be explained from the period length, which is higher than the high OCD band wavelengths, meaning that coupling between unit cells will be generally weak. However, it can also be seen that despite the homogeneity in the band's spectral position, there are significant inhomogeneities in the actual average OCD value along that band. For this reason, in Figure S7b the spectrally averaged OCD values within that band are plotted against the lattice period, with a visible maximum at $\Lambda = 1850$ nm. For this lattice period, the reflectance and transmittance spectra are represented in Figure S7c. In general, these spectra show high transmittances for all wavelengths, reaching values around 0.9 for the [1250, 1400] nm interval, and minima around 0.6-0.7 for the wavelengths where the anapolar modes are expected. This characteristic is also interesting since high transmittances, while not beneficial for CD enhancement, can lead to more precise measurements in practice⁸.

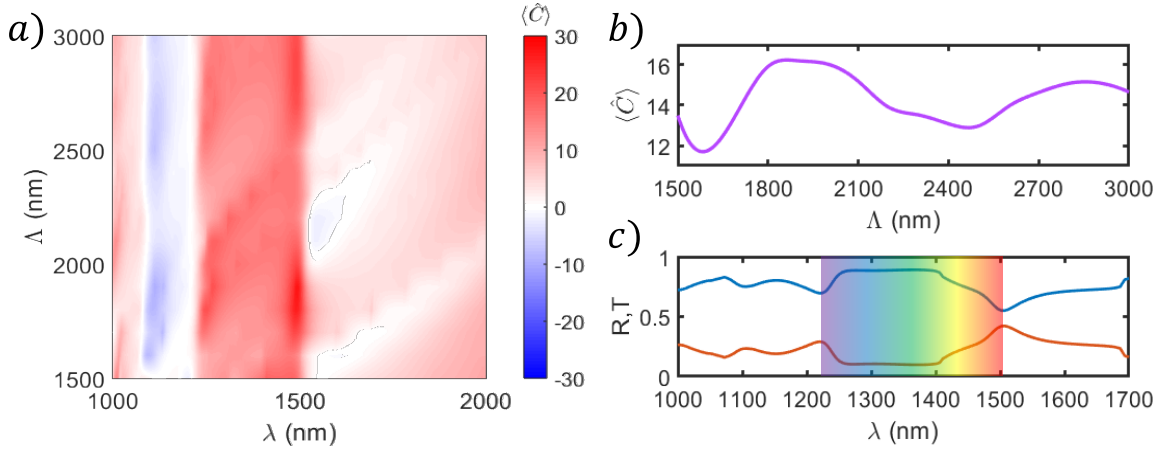


Figure S8. Average OCD in the inner gap of the structure. a) Calculated vs incident wavelength λ and lattice period Λ . b) Averaged both spatially (within the inner gap) and spectrally (in the 1200-1500 nm band) vs the lattice period Λ . c) Reflectance (orange) and transmittance (blue) spectrum for the optimal $\Lambda = 1850$ nm metasurface design, with the high OCD enhancement band in the background.

SVI. DETAILS OF CHIRAL FEM SIMULATIONS

As mentioned in the main text, the electromagnetic interaction of chiral media is mediated by the Pasteur parameter in the constitutive relations

$$\begin{aligned} \mathbf{D} &= \epsilon \mathbf{E} - i\kappa \sqrt{\epsilon_0 \mu_0} \mathbf{H} \\ \mathbf{B} &= \mu \mathbf{H} + i\kappa \sqrt{\epsilon_0 \mu_0} \mathbf{E} \end{aligned} \quad (\text{S1})$$

To perform numerical calculations, the built-in constitutive relations in COMSOL were effectively transformed into equations S1. The chiral medium, mimicking a real sample, was set to a constant refractive index $n = 1.45 - 0.01i$ and a complex Pasteur parameter $\kappa = 10^{-4} + 10^{-6}i$, exhibiting a large but still realistic magnitude, with a typical ratio between real and imaginary part⁹. To minimize numerical error, mirror-symmetric meshing in the simulation region was utilized¹⁰.

SVII. CIRCULAR DICHROISM AND CHIRALITY TRANSFER: THEORY AND SIMULATION

Simulation results in Figure 4 in the main text showcase that CD transmission enhancement is not proportional to the average OCD enhancement divided by the mean transmittance as predicted in other works¹¹. This equation holds when the differential reflectance in the metasurface ΔR is neglected in comparison with the differential transmittance ΔT . Then, a relation between ΔT and the differential absorbance ΔA can be established as

$$\Delta T = -\Delta A - \Delta R \simeq -\Delta A \quad (\text{S2})$$

where the differential absorbance can then be related to the chirality transfer mechanism and the OCD enhancement divided by the incident power¹²

$$CD_{Enh} = \frac{CD}{CD_0} = CD_{Enh,CT} + CD_{Enh,OCD} = \frac{\Delta P_{abs}}{\Delta P_{abs,0}} \frac{\langle C \rangle}{C_{inc}} \quad (\text{S3})$$

However, as Figure S9a showcases, differential reflectance and transmittance are similar in absolute value for all wavelengths, meaning that the contributions from OCD enhancement and chirality transfer can not be established following equation S3. In order to establish this contribution, additional FEM simulations were carried out, with a pure imaginary Pasteur parameter $\kappa = 10^{-3}i$, consistent with the optical modelling of chiral media in other works^{11,13,14}.

These results can be found in Figure S9b and c, which show that CD enhancement by simulation is far more similar to the OCD enhancement term in equation S3. Slight mismatches in amplitude and wavelength are found between theory and simulation. These can be attributed to substantial differential reflectance near the CD enhancement maximum.

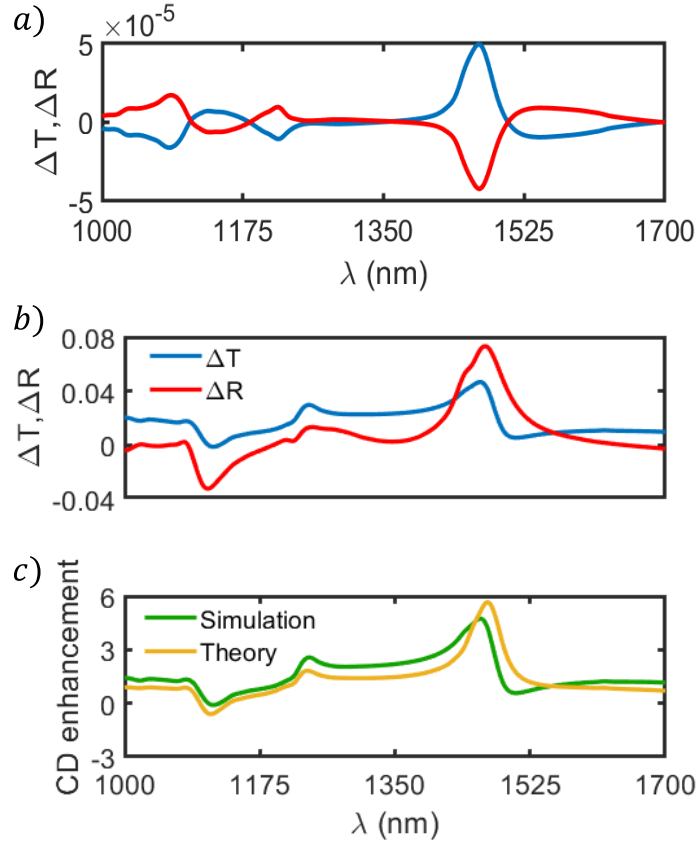


Figure S9. Differential transmittance (ΔT) and reflectance (ΔR) between circular polarizations in the $\Lambda = 1850$ nm metasurface with chiral medium characterized by a) $\kappa = 10^{-4} + 10^{-6}i$, and b) $\kappa = 10^{-3}i$. c) Simulation and theory results for CD enhancement with $\kappa = 10^{-3}$.

Since for the system with $\kappa = 10^{-4} + 10^{-6}i$ the terms in equation S3 cannot be explicitly separated due to the significant differential reflectance, chirality transfer contribution in Figure 4c was put together with ΔR as the difference between the simulation result and the OCD enhancement term.

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